1. Core Python Libraries Explained Simply

Digital Signal Processing in Python uses 4 fundamental libraries instead of physical lab instruments:

NumPy (`import numpy as np`)

The fast math engine. It handles large arrays of numbers simultaneously (Vectorization). Used to create time axes, calculate sines, compute FFTs, and evaluate sinc matrices.

SciPy (`import scipy`)

The scientific toolbox for advanced statistics, filtering, and signal processing routines.

Matplotlib (`import matplotlib.pyplot as plt`)

The visual oscilloscope screen. Draws continuous signals, stem plots for discrete samples, and frequency magnitude spectra.

Pandas (`import pandas as pd`)

The data table formatter. Organizes test results into clean tables with headers.

2. Key Functions & Code Breakdown

Signal Definition

```
def continuous_signal(t):
    return np.sin(2 * np.pi * 100 * t) + 0.5 * np.sin(2 * np.pi * 300 * t)
```

Generates two sinusoidal tones: $f_1 = 100\text{ Hz}$ (1.0 V) and $f_2 = 300\text{ Hz}$ (0.5 V). Maximum frequency $f_{\max} = 300\text{ Hz} \implies f_{\text{Nyquist}} = 600\text{ Hz}$.

Whittaker-Shannon Sinc Reconstruction

```
def sinc_reconstruction(sample_times, sample_values, t_fine, fs):
```

```
dt_matrix = t_fine[None, :] - sample_times[:, None]

sinc_matrix = np.sinc(fs * dt_matrix)

return np.dot(sample_values, sinc_matrix)
```

Implements $\hat{x}(t) = \sum x[n] \operatorname{sinc}(f_s(t - n T_s))$. `np.sinc` computes $\frac{\sin(\pi x)}{\pi x}$, and `np.dot` performs instant matrix multiplication to sum overlapping sinc curves.

Spectral Aliasing Calculation

When undersampling at $f_s = 400\text{ Hz}$ ($f_s < 600\text{ Hz}$):

$$f_{2,\text{alias}} = |f_2 - k f_s| = |300 - 1 \times 400| = 100\text{ Hz}$$

The 300 Hz tone folds back into the $[0, 200\text{ Hz}]$ band and appears at

100 Hz!

3. Summary Results Table

Sampling Case	Sampling Rate (f_s)	Ratio (f_s / f_{Nyquist})	MSE Error	Aliased f_2 Peak
Oversampled	1800 Hz	3.0x Nyquist	0.000005	300 Hz (No Alias)
Critically Sampled	600 Hz	1.0x Nyquist	0.123513	300 Hz (No Alias)
Undersampled	400 Hz	0.67x Nyquist	0.250021	**100 Hz (Aliased!)**

4. Predictive Viva Questions & Answers

> **Q1: What is the Nyquist-Shannon Sampling Theorem?**
>
> A signal with maximum frequency $f_{\max}$ can be perfectly reconstructed if sampled at $f_s \ge 2 f_{\max}$.

> **Q2: What is Aliasing?**
>
> High frequencies fold back into lower frequencies when $f_s < 2 f_{\max}$, distorting

> the reconstructed wave.

> **Q3: How do real systems prevent aliasing?**

> By placing an analog Low-Pass Anti-Aliasing Filter before the Analog-to-Digital Converter (ADC).

> **Q4: Why does sinc interpolation work?**

> The sinc pulse is the time-domain impulse response of an ideal brick-wall low-pass filter.

Prepared by Ciel for Apurba Maity • Cooch Behar Government Engineering College